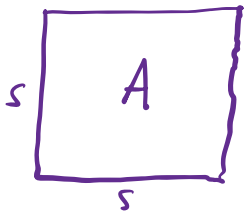
Problem 1. WORKED IN CLASS

A square's sides are growing at 3 cm/s. How fast is its **area** growing at the moment the side is 10 cm?

The method, every time.

1. **Draw** it, and label what moves with **letters** — not numbers.
2. Write what you **know** and what you **want**, as derivatives.
3. Find the **relating equation** — one equation, in those letters.
4. **Differentiate both sides with respect to t.**
5. **Now** substitute the instant's values, and solve.

(1)



(2) $\frac{ds}{dt} = 3$, $\frac{dA}{dt} = ?$ when $s = 10$

(3) $A = s^2$

(4) $\frac{d}{dt}A = \frac{d}{dt}(s^2) \rightarrow \frac{dA}{dt} = 2(s)^1 \frac{ds}{dt} = 2s \frac{ds}{dt}$ (10, 3)

(5) $\frac{dA}{dt} = 2(10)(3) = 60 \frac{cm^2}{sec}$